

NODERABBIT RUN-KEY PERSISTENCE

Exact registry-change hunt for the source-observed MicrosoftEdgeUpdate value launching a renamed Node.js runtime and script from the first NodeRabbit variant.

HUNTING QUERY

CONFIDENCE: MEDIUM

REPORT DATE

2 September 2026

STATUS

Untested implementation sketch

PRIMARY TELEMETRY

Defender XDR / DeviceRegistryEvents

ATT&CK

T1547.001

Campaign-specific persistence. A match is not proof of logon execution, payload integrity, C2 or compromise.

CONTROLLED EIGHT-PAGE THREAT DOSSIER

HKCU\RUN - NODEW.EXE - MSEDGE_UPDATE.JS

MICROSOFTEDGEUPDATE

HKCU\...CurrentVersion\Run



nodew.exe

msedge_update.js

EXACT VALUE CHAIN

hunt - collect - corroborate



EXECUTIVE ASSESSMENT

Decision-grade summary and strict evidence boundary.

ASSESSMENT

OPERATIONAL VALUE

9.3

Direct visibility into a source-observed user persistence chain that combines an exact Run value and two exact execution artifacts.

CLASSIFICATION

Campaign-specific persistence

Hunting query - Untested implementation sketch. The variant-specific values are precise but mutable.

3countries observed

3NodeRabbit variants

4exact query conditions

0production rules

Source fact

Variant 1 creates the HKCU Run value MicrosoftEdgeUpdate to execute nodew.exe msedge_update.js.

Analytical meaning

The combined registry location, value name and data tokens minimize overlap while preserving explainability.

Assumptions

Defender for Endpoint observed the modification and retained complete registry value data.

Promotion boundary

The first variant differs from later NodeRabbit builds. Do not generalize the artifact into a durable Node.js malware rule.



THREAT MECHANICS

Source-supported delivery and the observable registry boundary.

BEHAVIOR

01 Recruiter lure

A trojanized coding challenge is delivered through a job-search platform.

→

02 Package import

colorized_terminal launches a hidden Node.js implant.

→

03 AppData copy

The script is copied to Microsoft EdgeUpdate under the user profile.

04 Runtime clone

node.exe is cloned to nodew.exe and patched as GUI.

→

05 Run value

MicrosoftEdgeUpdate references the runtime and script.

→

06 Logon start

The registry mechanism is intended to relaunch the implant.

Behavior	Source support	Query coverage
HKCU Run persistence	Directly documented	Exact key suffix
MicrosoftEdgeUpdate value	Directly documented	Exact name
nodew.exe + msedge_update.js	Directly documented	Both required
Successful logon execution	Intended behavior	Not observed

ATT&CK

T1547.001 Registry Run Keys / Startup Folder directly applies. The query observes registry creation or modification, not subsequent execution.



TELEMETRY & EVIDENCE MODEL

Minimum data contract and observability gaps.

TELEMETRY

REQUIRED

DeviceRegistryEvents

Microsoft Defender for Endpoint registry creation or modification events with full value name and data.

OPTIONAL ENRICHMENT

DeviceFileEvents, DeviceProcessEvents, signer/hash reputation, logon activity, image collection and endpoint network telemetry.

Field	Reliability	Purpose
Timestamp, DeviceId, DeviceName	Documented	Scope and pivots
RegistryKey, RegistryValueName	Documented	Persistence location and value
RegistryValueData	Documented	Runtime and script tokens
ActionType	Documented; values in portal	Context only
InitiatingProcess*	Documented	Writer lineage and account
ReportId	Documented	Use with DeviceName and Timestamp

Unavailable through this query

Current registry state, payload content, runtime patching, successful logon execution, detached process, API traffic, encrypted C2 and actor attribution.

Coverage failure

The value predates onboarding, the change was not recorded, retention expired, RegistryValueData is incomplete, or a later variant uses a different mechanism.



HUNTING LOGIC

Four direct conditions with no threshold or correlation.

KQL

HYPOTHESIS

If an attacker deploys NodeRabbit variant 1 on Windows, registry telemetry should show a Run value named MicrosoftEdgeUpdate whose data references both nodew.exe and msedge_update.js. Legitimate overlap is reduced through the exact combined chain and resolved through writer, file, signer and process evidence.

KQL - UNTESTED HUNT

```
DeviceRegistryEvents
| where RegistryKey endswith @"\Software\Microsoft\Windows\CurrentVersion\Run"
| where RegistryValueName =~ "MicrosoftEdgeUpdate"
| where RegistryValueData contains "nodew.exe"
| where RegistryValueData contains "msedge_update.js"
| project Timestamp, DeviceId, DeviceName, ActionType,
    RegistryKey, RegistryValueName, RegistryValueType, RegistryValueData,
    PreviousRegistryValueData,
    InitiatingProcessFileName, InitiatingProcessFolderPath,
    InitiatingProcessCommandLine, InitiatingProcessSHA1, InitiatingProcessMD5,
    InitiatingProcessAccountName, InitiatingProcessAccountSid,
    InitiatingProcessIntegrityLevel, InitiatingProcessTokenElevation,
    IsInitiatingProcessRemoteSession, InitiatingProcessRemoteSessionIP, ReportId
| order by Timestamp desc
```

THRESHOLD	WINDOW	JOIN	KEY	STATUS
None	Analyst-selected	None	Exact value chain	Untested

Why Defender XDR

The documented registry schema exposes the key, value name, value data and initiating process directly without customer-specific parsing.



TRIAGE & INVESTIGATION

Turn a registry match into defensible endpoint evidence.

WORKFLOW

1

Validate the event

Confirm all four conditions and preserve Timestamp, DeviceName and ReportId.

2

Inspect the writer

Review initiating process, command line, account, integrity and remote-session context.

3

Collect referenced files

Locate nodew.exe and msedge_update.js; collect hashes and signatures.

4

Validate execution

Search for process starts at logon and establish parent, user and script path.

5

Scope the lure

Identify the coding-challenge archive, trojanized package and related endpoints.

Escalate

Matching files in user AppData, untrusted runtime, Node script execution or source-observed network evidence.

Benign context

Approved lab emulation or an internally documented updater with verified publisher and owner.

Inconclusive

Deleted value, missing files, partial data or historical artifact outside retention.

Response boundary

Do not attribute to Mirage Kitten from the registry event alone. Confirm the payload chain and scope before containment decisions.



FALSE POSITIVES & VALIDATION

What must be proven before operational use.

QUALITY CONTROL

SCENARIO	OVERLAP	DISPOSITION
Legitimate custom updater	L	Verify owner, publisher and deployment record
Security lab emulation	M	Scope approved test devices and window
Malware sample detonation	M	Confirm sandbox role and disposable image
User workstation match	H	Escalate for file and process corroboration

Validation steps

1. Verify registry-event retention and full value-data population.
2. Create an authorized benign Run value with the same structure.
3. Confirm key, value, data and initiating-process fields.
4. Run historical hunt and disposition every exact match.

Decision criteria

Success: controlled event appears and matches are dispositionable.

Rejection: legitimate overlap is material and cannot be separated.

Failure: registry event or value data is absent.

Tuning

Allowlist only verified device roles, publishers or deployment identities. Do not suppress the value name globally or remove the two value-data conditions.



CONCLUSION & REFERENCES

Coverage statement, cross-platform limits and document control.

FINAL

CONCLUSION

A precise persistence hunt, not a durable NodeRabbit detector.

The registry chain is directly source-supported and Defender fields are documented. Variant drift, environmental validation and execution evidence remain the main promotion barriers.

Cross-SIEM decision

Splunk: no reliable SPL without a verified EDR/registry source, index, sourcetype and exact key/value/data extractions.

Chronicle: no reliable YARA-L without a verified Windows-registry parser and exact UDM mapping.

Deferred ideas

Git-hook content detection, fake VS Code trust detection, path-only Node.js alerts, Azure Websites C2 domains, later scheduled-task variants and cross-platform translations.

References

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2. Microsoft Learn - DeviceRegistryEvents, updated 11 Jun 2025.
[learn\[.\]microsoft\[.\]com/en-us/defender-xdr/advanced-hunting-deviceregistryevents-table](#)
3. MITRE ATT&CK - T1547.001, modified 12 May 2026.
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Document control

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